

Particle-based model of supersaturation and turbulent mixing in atmospheric clouds

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April 3, 2026

Abstract

This work highlights the importance of supersaturation in the creation of atmospheric clouds. It aims at creating a minimalistic model in which the supersaturation and turbulent mixing can be studied in a controlled environment, later to compare it to theoretical predictions. Inside the chamber, temperature and water vapour mixing ratio is setup in such a way that would resemble air - cloud interface. A stochastic, particle - based model is created and investigated. Study showed that the saturation inside the cloud for optimised parameters reaches 0.012, and is lower in dry air surrounding it. The model developed will further be expanded to simulate condensation and formation of water droplets inside clouds.

1 Introduction

The technological importance of having reliable cloud models can't be understated. It lies in a foundation of weather forecasting, understating atmosphere dynamics and such need has been recognised for many years. Miniature simulations of cloud dynamics can give us insight into the behaviour of clouds under controlled environment giving physicist room to tweak and improve empirical models but also to test and correct theoretical results. This work walks through the development of a model calculating supersaturation inside a cloud. It will further be developed to take into account condensation and creation of water droplets.

1.1 Particles' movement

Every particle has predetermined set of parameters: position r , velocity v , temperature T and vapor mixing ratio Q . The stochastic time-step that governs the particles' movement in the developed model was based on work by G.C. Abade [3]. Every step the velocity update is proportional to v/τ and the fluctuations of velocity simulating the Brownian motion are approximated by Wiener process presented in eq. 1

$$\begin{cases} dv = -\frac{v}{\tau}dt + \sqrt{\frac{2\sigma_v^2}{\tau}}dW \\ dr = vdt \end{cases} \quad (1)$$

where v is the velocity in the previous step, τ is the Lagrangian velocity integral time-scale, $\sigma_v = \sqrt{2k/3}$ is the particle velocity variance and dW is the increment of the Wiener process defined as:

$$dW = \mathcal{N}(0,1)\sqrt{dt} \quad (2)$$

where $\mathcal{N}(0,1)$ is a random sample taken from a normal distribution. Usually in the literature, the particle is also a subject to statistical diffusion $dr = vdt + \sqrt{2\Gamma}dW$, however this term can be omitted due to characteristic time-scale of the simulation, as the turbulent diffusion has much greater effect on it [2].

1.2 Vapor mixing and supersaturation

Dry and hot air coming into a region of colder gas by diffusion causes supersaturation of water vapor inside that region. Supersaturated air then condenses into droplets in order to equalize the excess. Goal of this simulation is only concerned with researching the supersaturation itself given by equation (3).

$$S = \frac{e}{e(T)} - 1 \quad (3)$$

where

- S - saturation
- e - water vapor pressure
- e(T) - saturation vapor pressure

Saturation vapor pressure can be calculated using eq. (4)

$$e(T) = \varepsilon_0 \exp\left(\frac{L_{v0}}{R_v} \left(\frac{1}{T_0} - \frac{1}{T}\right)\right) \quad (4)$$

where T_0 , ε_0 - temperature and pressure of water at triple point, L_{v0} - latent heat of vaporization, R_v - specific gas constant for water vapor.

2 Particle model

2.1 Model parameters

Hyperparameters needed for the stability and physics of the model were mainly based on work by I. Kumela [2]. C_T, C_Q - velocity-to-scalar time scale ratios for temperature and mixing ratio, ω - characteristic turbulence frequency based on Kolmogorov hypothesis, L - length of the side of chamber, ε - turbulent kinetic energy dissipation per unit mass, $\tau = \frac{4k}{2C_0\varepsilon}$. Important parameters that have profound effect on the final shape of the saturation curve are initial T and Q. Optimal values were found by trial and error, and finally concluded to be: ($T_{in} = 293.16$ K, $Q_{in} = 1.1 \cdot \varepsilon(T_{in})$) and ($T_{out} = 283.16$, $Q_{out} = 0.6 \cdot \varepsilon(T_{out})$)

2.2 Temperature and vapor pressure mixing

Each particle in the developed model has two vector parameters - velocity v and position r and two scalar parameters - Temperature T and vapor mixing pressure Q . Kinematics of this system are governed by equations (1), with the inclusion of periodic boundary conditions on every wall. Two approaches of the parameter relaxation were studied: one in which the average temperatures and vapor mixing ratios were predetermined and fixed, and one in which the relaxation goes towards average of neighboring particles. In the first approach, based on particle's position in the chamber, T and Q are being relaxed to either (T_{in}, Q_{in}) or (T_{out}, Q_{out}). That setup is presented in fig. 1. Q is set up to be slightly different from equilibrium state corresponding to the temperature of the region to avoid anomalies after reaching stationary state (Hence the 0.6 and 1.1 factors). During each time step dt , the change in scalar variables is proportional to the difference between the value of predefined parameter space in which the particle currently is located and the parameter value of said particle (5, 6).

$$T_{new} = T_{old} - C_T \omega (T_{old} - \langle T \rangle) dt \quad (5)$$

$$Q_{new} = Q_{old} - C_Q \omega (Q_{old} - \langle Q \rangle) dt \quad (6)$$

Where variables inside brackets represent predefined parameters of the scalar field that the particles are currently in.

In the relaxation towards the average approach, the values in the bracket represent the average of parameters in the neighbourhood of a particle. The averages were calculated over the strips of height L and width $L/50$. We can simplify the definition of neighbourhood to these stripes instead of boxes thanks to periodic boundary conditions and translational symmetry along the y axis. After each step, saturation for each particle is calculated using eq. (3), and then is being averaged over stripes of dimension $(L/50, L)$.

All the integration, boundary conditions and parameter relaxation were realised using Python programming language and NumPy package for numerical computation. [5]

2.3 Homogeneity of the system

In order to ensure the *well-mixed condition* of the system, the distribution of particles needs to be uniform. In order to test that hypothesis, the particles were counted over $N \times N$ even squares. Based on equation (1) the fluctuations of number of particles should follow gaussian distribution with standard deviation equal to $\sqrt{N_{particles}/N_{squares}}$. Results of that test are presented in figure 3. One can see that in this setup (100k particles divided over 9 squares), the distribution behaves as expected.

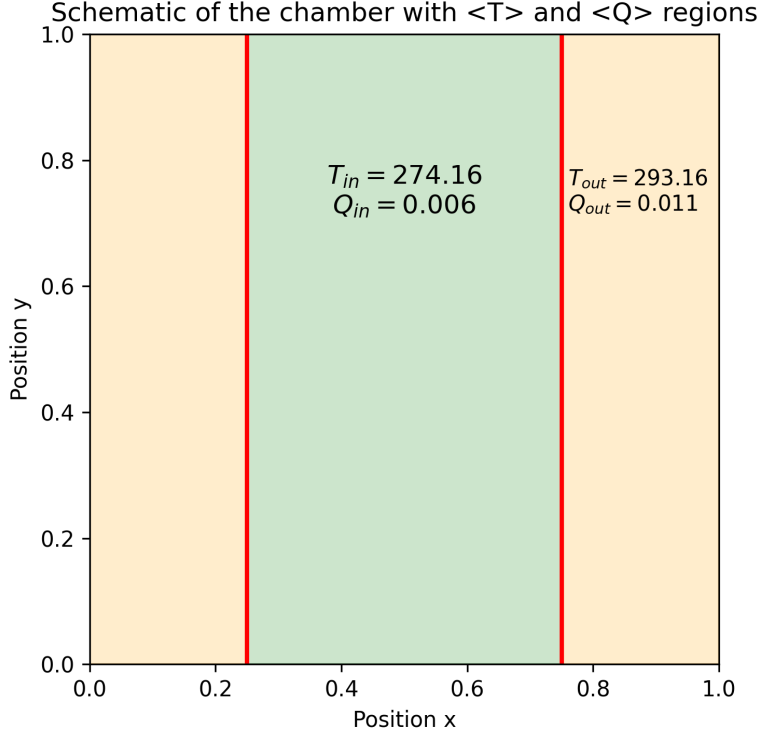


Figure 1: Definition of T and Q fields inside the chamber. Strip in the middle represents a slab of cloud with lower temperature than the surrounding air.

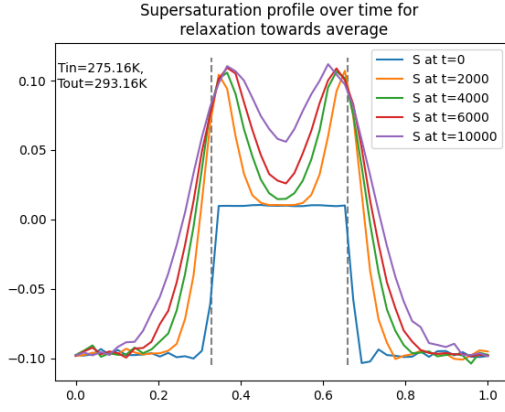
3 Conclusion

3.1 Fixed-parameters approach

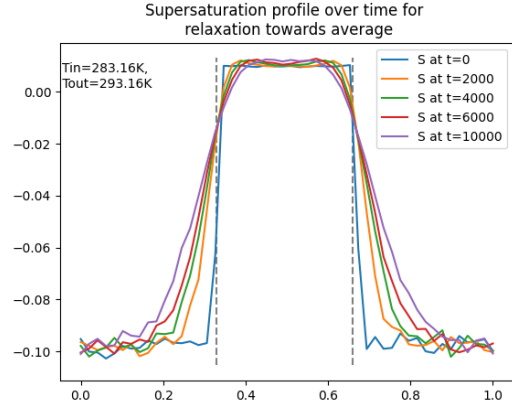
The fixed-parameter approach of the simulation gave very unstable and anomalous results. It was still an informative study, as the computation complexity was $O(N)$ and it sped up the simulation extensively, allowing for tests on much bigger number of particles (up to a million). The supersaturation profiles were very "stiff", as the slopes at the edge of the cloud were steep and non-physical. Also, for big differences in temperatures, the characteristic needles appeared (as seen in fig. 2d). Hence this approach was quickly abandoned for benefit of relaxation towards average.

3.2 Average fields approach

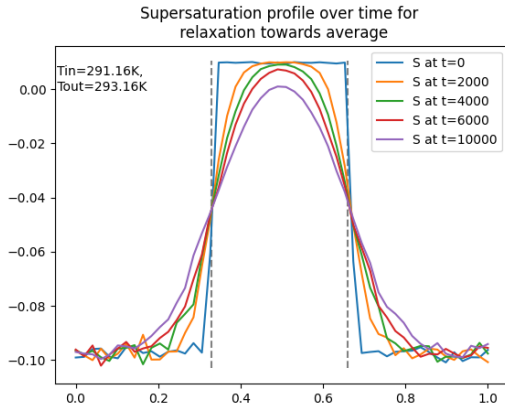
The second simulation produces supersaturation profiles more consistent with existing literature. Without the limitation of predetermined fields, the parameters of particles were more physical and free from anomalous needles. Simulation was run on 10000 particles over 10000 to 100000 steps (depending on the computation time which differs based on the initial temperatures). The shape of the supersaturation curve along x-axis follows results derived in [1], and is shown in figures (2 a-c). However, the values of saturation deeply depend on the initial parameters T_{in} and T_{out} . The following plots present the distribution for different parameters 2. The numerical values are presented in table 1. Based on the information in work by J.Kainz et al. [4], supersaturation follows literature values for the best-case scenario being $S_{in} \approx 0.01$, $S_{out} \approx -0.1$ the parameters chosen for development of further simulation are $C_Q = 10$, $C_T = 10$, $C_0 = 2.1$, $T_{in} = 283K$, $T_{out} = 293K$.



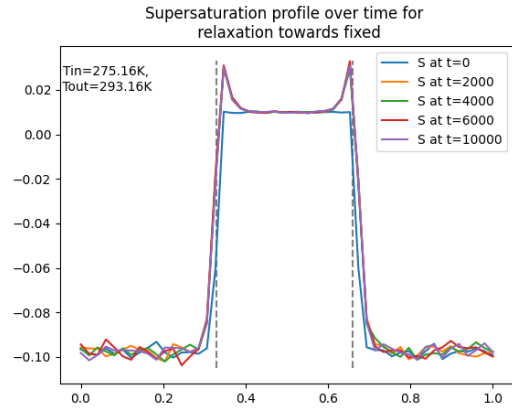
(a) $T_{in} = 273.16K$



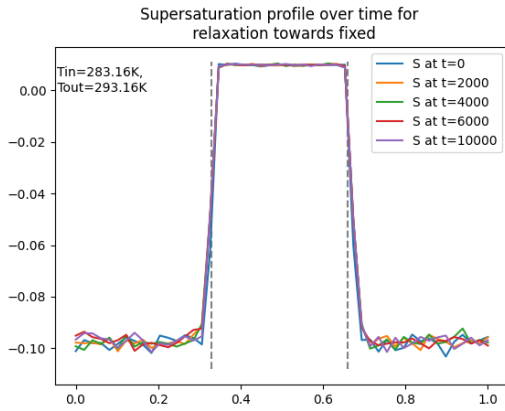
(b) $T_{in} = 283.16K$



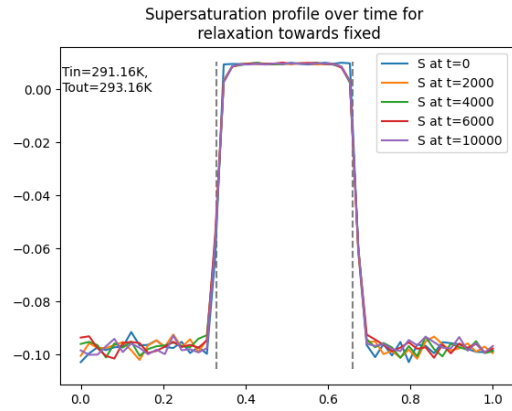
(c) $T_{in} = 291.16K$



(d) $T_{in} = 275.16K$



(e) $T_{in} = 283.16K$



(f) $T_{in} = 291.16K$

Figure 2: Profiles of the saturation in different timestamps for various T_{in} , with $T_{out} = 293.16$ for all simulations. Plots present both methods of relaxation: towards average field across x dimension (a,b,c) and towards fixed T and Q values (d,e,f).

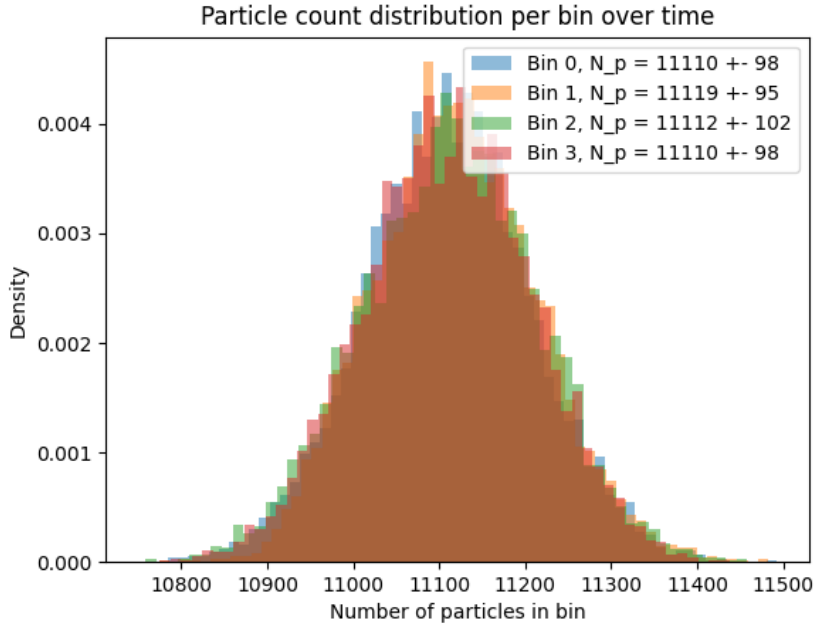


Figure 3: Distribution of number of particles in 9 squares after 10000 time steps of the simulation.

$T_{in}[C]$	$T_{out}[C]$	S_{in}	S_{out}
18	20	-0.0273	-0.0948
10	20	0.0012	-0.0929
2	20	0.1017	-0.0873

Table 1: Saturation averages inside and outside the cloud for a model with relaxation towards average

References

- [1] J. Fries et al. *Lagrangian Supersaturation Fluctuations at the Cloud Edge* DOI: 10.1103/Phys-RevLett.131.254201
- [2] I. Kumela *Minimalistic particle-based model of a cloud chamber*
- [3] G.C. Abade *Persistent mixed-phase states in adiabatic cloud parcels under idealised conditions* DOI: 10.1002/qj.4775
- [4] J. Kainz et al. *Benchmarking Turbulence Models to Represent Cloud-Edge Mixing* DOI 10.1175/JAS-D-24-0228.1.
- [5] <https://github.com/Piotr-Marczak13/cloud-physics.git>